

Time Series Analysis

Chapter 1 — Basic statistical concepts

Aymen Ben Brik
aymen.benbrik@esprit.tn
Esprit School of Business

2025-2026

Contents

1 Basic statistical concepts	1
Introduction	1
1.1 Moments of a distribution	1
1.2 Classical estimators and sampling distributions	2
1.2.1 Sample mean and sample variance	2
1.2.2 Sampling distribution under normality	2
1.2.3 Asymptotic results	3
1.3 Estimation methods	3
1.3.1 Method of Moments (MoM)	3
1.3.2 Maximum Likelihood Estimation (MLE)	4
1.4 Multivariate data	4
1.4.1 Joint, marginal, and conditional distributions	4
1.4.2 Covariance and correlation	5
1.5 Statistical inference	5
1.5.1 Confidence intervals for the mean	5
1.5.2 Hypothesis testing	5
Summary	6

Chapter 1

Basic statistical concepts

Introduction

Before studying time series proper, this chapter reviews the statistical concepts on which the rest of the course relies: *moments* of a distribution, classical *point estimators*, *Method of Moments* and *Maximum Likelihood Estimation*, sampling distributions, and basic *inference* (confidence intervals and hypothesis tests).

The convention throughout the chapter is $X_1, \dots, X_n$ are *independent and identically distributed* (i.i.d.) draws from a distribution with density $f(x; \theta)$ parameterised by some unknown θ . We observe a sample and want to learn θ . Time series will later violate the i.i.d. assumption, but the basic estimators introduced here remain the building blocks.

1.1 Moments of a distribution

Definition 1 – Raw and central moments

Let X be a real-valued random variable with density f and mean $\mu = \mathbb{E}[X]$.

- The *raw ℓ -th moment* is $m'_\ell = \mathbb{E}[X^\ell] = \int x^\ell f(x) dx$.
- The *central ℓ -th moment* is $m_\ell = \mathbb{E}[(X - \mu)^\ell]$.

Useful quantities.

$$\begin{aligned} \mu &= m'_1, & \sigma^2 &= m_2 = \text{Var}[X], \\ \text{skewness } S(X) &= \frac{m_3}{\sigma^3}, & \text{kurtosis } K(X) &= \frac{m_4}{\sigma^4}. \end{aligned}$$

Remark – Interpretation

- $S(X)$ measures *asymmetry*. $S = 0$ for any symmetric distribution; $S > 0$ for a right-skewed distribution.
- $K(X)$ measures *tail-heaviness*. The Gaussian has $K = 3$; the *excess kurtosis* $K - 3$ is therefore the standard comparison: $K - 3 > 0$ indicates heavier tails than the Gaussian (leptokurtic).

Example 2 – Moments of the standard normal

For $X \sim \mathcal{N}(\mu, \sigma^2)$ one shows by direct integration or via the moment-generating function:

$$\mathbb{E}[X] = \mu, \quad \text{Var}[X] = \sigma^2, \quad m_3 = 0, \quad m_4 = 3\sigma^4.$$

Hence $S(X) = 0$ and $K(X) = 3$.

1.2 Classical estimators and sampling distributions**1.2.1 Sample mean and sample variance****Definition 3 – Empirical estimators**

Given i.i.d. data $X_1, \dots, X_n$:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Proposition 4 – Unbiasedness of $\bar{X}_n$ and S_n^2

For i.i.d. data with finite second moments, $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathbb{E}[S_n^2] = \sigma^2$.

Proof. For the mean: $\mathbb{E}[\bar{X}_n] = \frac{1}{n} \sum_i \mathbb{E}[X_i] = \mu$ by linearity. For the variance, expand:

$$\sum_i (X_i - \bar{X}_n)^2 = \sum_i X_i^2 - n \bar{X}_n^2.$$

Take expectations: $\mathbb{E}[X_i^2] = \mu^2 + \sigma^2$ and $\mathbb{E}[\bar{X}_n^2] = \mu^2 + \sigma^2/n$. Hence $\mathbb{E}[\sum_i (X_i - \bar{X}_n)^2] = n(\mu^2 + \sigma^2) - n(\mu^2 + \sigma^2/n) = (n-1)\sigma^2$. The $n-1$ denominator in S_n^2 ensures $\mathbb{E}[S_n^2] = \sigma^2$. $\square$

Remark – Why $n-1$?

Replacing the unknown μ by $\bar{X}_n$ in the variance formula “costs” one degree of freedom. Dividing by n would systematically underestimate σ^2 .

1.2.2 Sampling distribution under normality**Theorem 5 – Cochran-type results**

If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then:

1. $\bar{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$;
2. $\frac{(n-1)S_n^2}{\sigma^2} \sim \chi_{n-1}^2$;
3. $\bar{X}_n$ and S_n^2 are independent;
4. $\frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} \sim t_{n-1}$.

Remark

Result (4) follows from (1)–(3): the numerator is standard normal, the denominator is the square root of an independent χ_{n-1}^2 divided by its degrees of freedom — the very definition of Student's t .

1.2.3 Asymptotic results**Theorem 6 – Law of Large Numbers**

For i.i.d. data with $\mathbb{E}[X_i] = \mu$:

$$\bar{X}_n \xrightarrow{p} \mu \quad \text{as } n \rightarrow \infty.$$

Theorem 7 – Central Limit Theorem

For i.i.d. data with $\mathbb{E}[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

Remark – Practical consequence

Even when the X_i are not Gaussian, the sample mean behaves approximately as a normal random variable for large n . The rule of thumb $n \geq 30$ is conservative; in practice the convergence is faster for symmetric, light-tailed distributions and slower for skewed or heavy-tailed ones.

1.3 Estimation methods**1.3.1 Method of Moments (MoM)****Definition 8 – Method of Moments**

Let $\theta \in \mathbb{R}^k$. The MoM estimator $\hat{\theta}_n^{\text{MoM}}$ solves the system of equations

$$\mathbb{E}_\theta[X^p] = \frac{1}{n} \sum_{i=1}^n X_i^p, \quad p = 1, 2, \dots, k.$$

That is, theoretical moments are equated to sample moments.

Example 9 – $\mathcal{N}(\mu, \sigma^2)$ via MoM

With $\theta = (\mu, \sigma^2)$ and $k = 2$:

$$\begin{aligned} \mu &= \bar{X}_n, \\ \mu^2 + \sigma^2 &= \frac{1}{n} \sum_i X_i^2. \end{aligned}$$

Subtracting yields $\hat{\sigma}_{\text{MoM}}^2 = \frac{1}{n} \sum (X_i - \bar{X}_n)^2$, which differs from S_n^2 by a factor $n/(n-1)$.

1.3.2 Maximum Likelihood Estimation (MLE)

Definition 10 – Likelihood and MLE

For i.i.d. data with density $f(x; \theta)$:

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta), \quad \ell(\theta) = \log L(\theta).$$

The MLE is $\hat{\theta}_n^{\text{MLE}} = \arg \max_{\theta} \ell(\theta)$.

Example 11 – $\mathcal{N}(\mu, \sigma^2)$ via MLE

The log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2.$$

Setting $\partial \ell / \partial \mu = 0$ gives $\hat{\mu}_{\text{MLE}} = \bar{X}_n$. Then setting $\partial \ell / \partial \sigma^2 = 0$ gives $\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum (X_i - \bar{X}_n)^2$. So MLE coincides with MoM here, and both are biased for σ^2 (their expectation is $\frac{n-1}{n} \sigma^2$).

Theorem 12 – Asymptotic normality of the MLE

Under regularity conditions (identifiability, smoothness, finite Fisher information $I(\theta_0) > 0$):

$$\sqrt{n} (\hat{\theta}_n^{\text{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}).$$

Remark – MoM vs. MLE

- MoM is computationally simple — just match moments.
- MLE is *asymptotically efficient*: its variance attains the Cramér–Rao lower bound $I(\theta_0)^{-1}/n$.
- Both estimators are consistent (under regularity).
- For “nice” families (Gaussian, exponential, Poisson), MoM and MLE often coincide or are very close.

1.4 Multivariate data

1.4.1 Joint, marginal, and conditional distributions

Definition 13 – Joint and marginal density

For two random variables X, Y with joint density $f(x, y)$:

$$f_X(x) = \int f(x, y) dy, \quad f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)}.$$

The chain rule reads $f(x, y) = f_{Y|X}(y | x) f_X(x)$.

Definition 14 – Independence

X and Y are *independent* iff $f(x, y) = f_X(x) f_Y(y)$ for all x, y .

1.4.2 Covariance and correlation**Definition 15 – Covariance and correlation**

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)], \quad \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}X \text{Var}Y}}.$$

Proposition 16 – Sample versions

$$\widehat{\text{Cov}}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}), \quad \widehat{\text{Corr}}(X, Y) = \frac{\widehat{\text{Cov}}(X, Y)}{\widehat{\sigma}_X \widehat{\sigma}_Y}.$$

The sample correlation lies in $[-1, 1]$.

Warning – Correlation captures only linear dependence

$\text{Corr}(X, Y) = 0$ does not imply independence. Counter-example: let X be symmetric around 0 and $Y = X^2$. Then $\mathbb{E}[XY] = \mathbb{E}[X^3] = 0 = \mathbb{E}[X]\mathbb{E}[Y]$, hence $\text{Cov}(X, Y) = 0$, yet Y is a deterministic function of X .

1.5 Statistical inference**1.5.1 Confidence intervals for the mean****Definition 17 – Confidence interval**

A $(1 - \alpha)$ -confidence interval for μ is a random interval $[\widehat{L}_n, \widehat{U}_n]$ such that $\mathbb{P}(\widehat{L}_n \leq \mu \leq \widehat{U}_n) = 1 - \alpha$ in repeated sampling.

Gaussian sample, known σ (z -interval). Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ known. By Theorem 1.2.2 (1):

$$\text{CI}_{1-\alpha}(\mu) = \left[\bar{X}_n - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

Gaussian sample, unknown σ (t -interval). Replace σ by S_n and use Student's t (Theorem 1.2.2 (4)):

$$\text{CI}_{1-\alpha}(\mu) = \left[\bar{X}_n - t_{n-1, 1-\alpha/2} \frac{S_n}{\sqrt{n}}, \bar{X}_n + t_{n-1, 1-\alpha/2} \frac{S_n}{\sqrt{n}} \right].$$

1.5.2 Hypothesis testing**Definition 18 – Test on the mean**

Two-sided test: $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$. Test statistic

$$T = \frac{\bar{X}_n - \mu_0}{S_n / \sqrt{n}} \sim t_{n-1} \text{ under } H_0.$$

Reject H_0 at level α if $|T| > t_{n-1, 1-\alpha/2}$, i.e. the p -value $2 \cdot \mathbb{P}(T_{n-1} > |T|) < \alpha$.

Remark – Errors

- Type I error (size of the test): $\alpha = \mathbb{P}(\text{reject } H_0 \mid H_0)$.
- Type II error: $\beta = \mathbb{P}(\text{not reject } H_0 \mid H_1)$.
- *Power* of the test: $1 - \beta$, the probability of correctly rejecting H_0 when H_1 is true.

Example 19 – Numerical illustration

Suppose $n = 50$ Atlanta monthly average temperatures with sample mean $\bar{X}_{50} = 61.2$ and sample standard deviation $S_{50} = 16.8$. Test $H_0 : \mu = 60$ against $H_1 : \mu \neq 60$ at $\alpha = 0.05$.

$$T = \frac{61.2 - 60}{16.8/\sqrt{50}} \approx 0.505, \quad t_{49,0.975} \approx 2.010.$$

Since $|T| < 2.010$, we do *not* reject H_0 ; $p\text{-value} \approx 2 \cdot \mathbb{P}(t_{49} > 0.505) \approx 0.616$. The corresponding 95% CI is $[61.2 - 4.78, 61.2 + 4.78] = [56.4, 66.0]$, which contains $\mu_0 = 60$, consistent with the test conclusion.

Warning – Common pitfalls

- “Failing to reject H_0 ” is *not* “proving H_0 ”.
- A small p -value does not imply a large effect — it only signals statistical significance, not practical importance.
- Multiple testing inflates Type I error: use Bonferroni or FDR corrections when running many tests.

Summary

Quantity	Estimator	Sampling distribution (Gaussian data)
μ	$\bar{X}_n = \frac{1}{n} \sum X_i$	$\mathcal{N}(\mu, \sigma^2/n)$
σ^2	$S_n^2 = \frac{1}{n-1} \sum (X_i - \bar{X}_n)^2$	$(n-1)S_n^2/\sigma^2 \sim \chi_{n-1}^2$
Pivot	$T = (\bar{X}_n - \mu)/(S_n/\sqrt{n})$	t_{n-1}

What to remember.

- Two estimation routes: MoM (match moments) and MLE (maximise likelihood). Both consistent; MLE asymptotically efficient.
- Under normality, the sampling distribution of $\bar{X}$ is exactly Gaussian; the CLT extends this asymptotically.
- Confidence intervals and hypothesis tests on the mean rely on Student’s t when σ is unknown.